

Final Project

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1 Problem 1: Cart-Pole with Actuator Dynamics: DDP and iLQR

Prerequisite: cart-Pole Dynamics

The dynamic equation of cart-pole system is as follows:

$$\ddot{x} = \frac{\ell m_p \sin(\theta) \dot{\theta}^2 + F + m_p g \cos(\theta) \sin(\theta)}{m_c + m_p (1 - \cos^2(\theta))} \quad (1.1)$$

$$\ddot{\theta} = -\frac{\ell m_p \cos(\theta) \sin(\theta) \dot{\theta}^2 + F \cos(\theta) + (m_c + m_p) g \sin(\theta)}{\ell m_c + \ell m_p (1 - \cos^2(\theta))} \quad (1.2)$$

(a)

Denote $\mathbf{x} = [x, \theta, \dot{x}, \dot{\theta}, F]$ as the augmented state vector. The dynamics of cart-pole system can be written as follows:

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{x} \\ \dot{\theta} \\ \ddot{x} \\ \ddot{\theta} \\ \dot{F} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{\theta} \\ \frac{\ell m_p \sin(\theta) \dot{\theta}^2 + F + m_p g \cos(\theta) \sin(\theta)}{m_c + m_p (1 - \cos^2(\theta))} \\ -\frac{\ell m_p \cos(\theta) \sin(\theta) \dot{\theta}^2 + F \cos(\theta) + (m_c + m_p) g \sin(\theta)}{\ell m_c + \ell m_p (1 - \cos^2(\theta))} \\ \frac{u - F}{\tau_a} \end{bmatrix} = \mathbf{f}(\mathbf{x}, u) \quad (1.3)$$

- Forward Euler Discretization

Given the continuous-time nonlinear dynamics

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u), \quad (1.4)$$

the Forward Euler method with sampling time h approximates the state evolution as

$$\mathbf{x}_{k+1} = \mathbf{x}_k + h \mathbf{f}(\mathbf{x}_k, u_k). \quad (1.5)$$

For the augmented cart-pole state the discrete-time dynamics become

$$x_{k+1} = x_k + h \dot{x}_k, \quad (1.6)$$

$$\theta_{k+1} = \theta_k + h \dot{\theta}_k, \quad (1.7)$$

$$\dot{x}_{k+1} = \dot{x}_k + h \ddot{x}_k, \quad (1.8)$$

$$\dot{\theta}_{k+1} = \dot{\theta}_k + h \ddot{\theta}_k, \quad (1.9)$$

$$F_{k+1} = F_k + h \frac{u_k - F_k}{\tau_a}. \quad (1.10)$$

Therefore, the discrete nonlinear dynamics can be written compactly as

$$\mathbf{x}_{k+1} = F_d(\mathbf{x}_k, u_k) = \mathbf{x}_k + h \mathbf{f}(\mathbf{x}_k, u_k). \quad (1.11)$$

As an alternative, a higher-order numerical integration method such as the fourth-order Runge–Kutta (RK4) scheme may be used to improve integration accuracy.

(b)

Define $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]$ and $\mathbf{U} = [u_0, \dots, u_{N-1}]$ as matrices of state and control trajectory. Then a cost function for the swing-up task is as follows:

$$J(\mathbf{X}, \mathbf{U}) = \frac{1}{2}(\mathbf{x}_N - \mathbf{x}_{ref})^T \mathbf{Q}_N (\mathbf{x}_N - \mathbf{x}_{ref}) + \frac{1}{2} \sum_{k=0}^{N-1} ((\mathbf{x}_k - \mathbf{x}_{ref})^T \mathbf{Q} (\mathbf{x}_k - \mathbf{x}_{ref}) + Ru_k^2) \quad (1.12)$$

where

$$l_N(\mathbf{x}_N) = \frac{1}{2}(\mathbf{x}_N - \mathbf{x}_{ref})^T \mathbf{Q}_N (\mathbf{x}_N - \mathbf{x}_{ref}) \quad (1.13)$$

is the terminal cost and

$$\sum_{k=0}^{N-1} l_k(\mathbf{x}_k, u_k) = \frac{1}{2} \sum_{k=0}^{N-1} ((\mathbf{x}_k - \mathbf{x}_{ref})^T \mathbf{Q} (\mathbf{x}_k - \mathbf{x}_{ref}) + Ru_k^2) \quad (1.14)$$

is the running cost. Define the value function:

$$V(\mathbf{x}_k) = \min_{u_k, u_{k+1}, \dots, u_{N-1}} \left(\sum_{t=k}^{N-1} l_t(\mathbf{x}_t, u_t) + l_N(\mathbf{x}_N) \right) \quad (1.15)$$

The Bellman optimality condition provides a recursion on $V(\mathbf{x}_k)$ as follows:

$$V(\mathbf{x}_k) = \min_{u_k} (l_k(\mathbf{x}_k, u_k) + V(\mathbf{x}_{k+1})) \quad (1.16)$$

$$= \min_{u_k} (l_k(\mathbf{x}_k, u_k) + V_{k+1}(F_d(\mathbf{x}, u))) \quad (1.17)$$

(c)

Define the local action-value function $Q(\delta \mathbf{x}_k, \delta u_k)$ as follows:

$$Q(\delta \mathbf{x}_k, \delta u_k) = l_k(\mathbf{x}_k + \delta \mathbf{x}_k, u_k + \delta u_k) + V_{k+1}(F_d(\mathbf{x}_k + \delta \mathbf{x}_k, u_k + \delta u_k)) \quad (1.18)$$

Then we can approximate $Q(\delta \mathbf{x}_k, \delta u_k)$ in a quadratic form by expanding to 2-nd order of its Taylor expansion:

$$\begin{aligned} Q(\delta \mathbf{x}_k, \delta u_k) \approx Q(\mathbf{0}, 0) + Q_{x,k}^T \delta \mathbf{x} + Q_{u,k}^T \delta u_k + \frac{1}{2} (\delta \mathbf{x}_k^T Q_{xx,k} \delta \mathbf{x}_k + \delta u_k^T Q_{uu,k} \delta u_k) \\ + \delta u_k^T Q_{ux,k} \delta \mathbf{x}_k \end{aligned} \quad (1.19)$$

where from definition of 1.18 we can determine:

$$\begin{aligned} Q_k(0, 0) &= l(\mathbf{x}_k, u_k) + V_{k+1}(\mathbf{x}_{k+1}), \\ Q_x &= l_x + F_{dx}^T V_{x,k+1}, \\ Q_u &= l_u + F_{du}^T V_{x,k+1}, \\ Q_{xx} &= l_{xx} + F_{dx}^T V_{xx,k+1} F_{dx} + V_{x,k+1}^T F_{dxx}, \\ Q_{uu} &= l_{uu} + F_{du}^T V_{xu,k+1} F_{du} + V_{x,k+1}^T F_{duu}, \\ Q_{ux} &= l_{ux} + F_{du}^T V_{xx,k+1} F_{dx} + V_{x,k+1}^T F_{dux}. \end{aligned}$$

where the subscription $k+1$ indicate the value taken at time $k+1$. Otherwise we evaluate at time k .

(d)

Now we minimize $Q(\delta \mathbf{x}_k, \delta u_k)$ in 1.19 over δu_k , we have

$$\delta u_k = \arg \min_{\delta u_k} Q(\delta \mathbf{x}_k, \delta u_k) = \arg \min_{\delta u_k} \frac{1}{2} \delta u_k^T Q_{uu,k} \delta u_k + Q_{u,k}^T \delta u_k + \delta u_k^T Q_{ux,k} \delta \mathbf{x}_k \quad (1.20)$$

$$= -Q_{uu}^{-1} \quad (1.21)$$